Adaptive Scheduling under Uncertain Processing Times

Research summary | AI + Operations Research

Research motivation

How do forecast updates and replanning triggers affect realized job-shop scheduling under controlled information and optimization budgets? Improving a forecast need not improve the resulting execution.

Technical approach

Independent Python discrete-event simulator with exact Fraction timing; private scenario/public observation boundary; residual OR-Tools CP-SAT with frozen running work; safe plan commitment and dispatch; fixed nominal and machine-level EMA forecasts (alpha 0.25); periodic, event and forecast-threshold triggers.

Controlled pilot

64 final runs: tiny diagnostic plus public abz5, four duration regimes, seeds 11/29, one optimizer repetition. Paired policies use identical exogenous manifests and configured effort caps. Recorded start-time and authorizing-plan-time predictions are evaluated separately; fallback coverage is explicit.

Evidence-based findings

Stationary abz5 EMA-periodic versus nominal-periodic changes makespan by -161 and +159 across the two seeds. Event-triggered B3 starts 80.0% of operations through fallback after budget pressure; its outcomes cannot be treated as pure trigger effects. Threshold triggering has favorable and unfavorable cases. No general superiority or statistical significance claim is made.

Research skills demonstrated

Python engineering; nonpreemptive scheduling and CP-SAT modeling; discrete-event simulation; online calibration; experimental controls; adversarial tests; independent metric reconstruction; solver-status, budget and provenance accounting; scientific communication.

Repository and reproducibility

130 regression tests pass. Final raw scenarios, execution ledgers, solver responses, plan forecasts and reproducible figures are supplied. README contains tested smoke, audit and plotting commands. Source-only portfolio is prepared for review; no public repository URL or publication is claimed.

Scope and next step

Independent research portfolio for AI + Operations Research RA applications, including HKU. No HKU affiliation or supervision is implied. Main limits: one public instance, few seeds, fallback confounding, policy-dependent prediction exposure and platform-sensitive reoptimization. Next: externally reviewed preregistration for a broader study.